

Communication Protocol

V4.0

1. Revision History

No	Description	Date	Version	Author
1.	Initial Release	2019.06.11	V1.0	
2.	Revision	2023.06.12	V4.0	

1. Basic Information

1.1 Terminology

Abbreviation	Full Name
MC	Motor Controller
BMS	Battery Management System
DU	Display Unit
UART	Universal Asynchronous Receiver/Transmitter

2. Definition

This protocol provides basic information describing how the Motor Controller, Battery Management System, Display Unit, and service module are connected for communication.

3. Functional Protocol

3.1 Definition

This section describes the internal data structure of the BMS that can be read by the controller via UART, RS485, or CAN.

3.2 Data Type Definitions

Data Type	Description	Length (bytes)	Min Value	Max Value
uint4	Unsigned long integer	4	0	4294967295
sint4	Signed long integer	4	-2147483648	2147483647
uint2	Unsigned integer	2	0	65535
sint2	Signed integer	2	-32768	32767
uint1	Unsigned char	1	0	255
sint1	Signed char	1	-128	127
bool1	Boolean	1	0	1
textn	Text string or byte string	n	n/a	n/a

3.4 Access Rights Definition

Data access rights are defined as follows:

- ro: Read only
- rw: Read Write
- wo: Write Only
- const: Constant

4. Physical Layer

4.1 UART

1. Physical Interface	UART	
2. Baud Rate	9600 bps	
3. Communication Format	9600 , N , 8 , 1	
4. Signal Level	TXD (Transmit)	“0”: <0.5V
		“1”: OC (Open Collector) (Voltage should be lower than 100V)
	RXD (Receive)	“0”: <0.5V
		“1”: >3V (Voltage should be lower than 100V)

4.2 CAN

1. Physical Interface	CAN
2. Baud Rate	250K

5.Communication Format

5.1 Basic Timing

All messages are sent by the host. All slave devices, upon receiving a message, check if the slave address matches. Data is only returned to the host if the slave address matches.

5.2 Address Assignment

Module	Address
BMS Master	0x01
Bluetooth Mobile App	0x80
GPRS	0x20
Upper Computer/Host Computer	0x40
Broadcast	0xFF

5.3 UART Communication Format

5.3.1 Upper Computer Sends Host Command

Start Flag	Upper Computer	Data ID	Data Length	Data Content	Checksum
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	Address (Comm. Module Address)				(1 byte)
0xA5 (Fixed)	0x80(UPPER-Add)	See Section 6	8 Bytes (Fixed)		

5.3.2 Slave Responds to Host Command

Start Flag	Comm. Module Address	Data ID	Data Length	Data Content	Checksum (1 byte)
0xA5(固定)	0x01(BMS-Add)	See Section 6	8 Bytes (Fixed)		

Note:

1. Each data item has a fixed length; two data items cannot be read at once.
2. The checksum is the sum of all preceding data (low byte only).
3. Example Send: 0xA5 0x40 0x90 0x08 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x00 0x7D. Example Return: 0xA5 0x01 0x90 0x08 0x02 0xFC 0x00 0x00 0x75 0xA1 0x00 0x00 0x52.

5.4 CAN Communication Format

5.4.1 Upper Computer Sends Host Command

CAN ID 4byte	Data Content
Priority + Data ID + BMS Address + Upper Computer Address (0x18100140)	8 bytes

5.4.2 BMS Responds to Host Command

CAN ID 4byte	Data Content
Priority + Data ID + Upper Computer Address + BMS Address (0x18104001)	8 bytes

Example Send: ID: 0x18900140, data:0x01 0x02 0x03 0x04 0x05 0x06 0x07 0x08

Example Return: ID: 0x18904001, data:0x01 0xbe 0x00 0x00 0x75 0x30 0x02 0x07

6. Communication Content Information

Message	Message ID	UPPER - BMS	Remark
Read Battery Capacity, Voltage	0x50	Send	Byte0~Byte7: Reserved
		Return	Byte0-Byte3: Design Capacity (mAh) Byte4-Byte7: Design Voltage (mV)
Read BMU, Cell Voltage, Temperature Count	0x51	Send	Byte0~Byte7:Reserved
		Return	byte0: Number of acquisition boards. byte1: Cell count of acquisition board 1. byte2: Cell count of acquisition board 2. byte3: Cell count of acquisition board 3. byte4: Temperature count of acquisition board 1. byte5: Temperature count of acquisition board 2. byte6: Temperature count of acquisition board 3.
Read Total Charge/Discharge Ah	0x52	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte3: Total Charge Ah (Ah). Byte4~byte7: Total Discharge Ah (Ah).
Read Battery Operation Mode	0x53	Send	Byte0~Byte7: Reserved
		Return	Byte0: Battery Type (0: LFP, 1: Ternary, 2: LTO). Byte1: Operation Mode (0x01: Long press to power on/off, 0x02: Short press to power on/off, others: omitted). Byte2: Year (2000 Offset). Byte3: Month Byte4: Day Byte5~byte6: Auto sleep time (S). Byte 7: Zero-drift current (0.1A)
Read Firmware Index Number	0x54	Send	Byte0~Byte7:Reserved
		Return	8 bytes ASC II code, identifying firmware version number. This number is used to locate the appropriate version for each different battery model.
Read Manufacturer Name	0x55	Send	Byte0~Byte7:Reserved
		Return	16byte, sent in 3 frames. Byte0: Frame number, starting from 1. Byte1~byte7: data

Read Battery Name	0x56	Send	Byte0~Byte7: Reserved
		Return	32byte, sent in 5 frames. Byte0: Frame number, starting from 1. Byte1~byte7: data
Read Battery Serial Number	0x57	Send	Byte0~Byte7: Reserved
		Return	32 bytes, sent in 5 frames. Byte0: Frame number, starting from 1. Byte1~byte7: data
Read Battery Production Date	0x58	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte7: Reserved
Read Cell Voltage Alarm	0x59	Send	Byte0~Byte7:Reserved
		Return	Byte0-byte1: Cell Overvoltage Level 1 Alarm (mV). Byte2-byte3: Cell Overvoltage Level 2 Alarm (mV). Byte4-byte5: Cell Undervoltage Level 1 Alarm (mV). Byte6-byte7: Cell Undervoltage Level 2 Alarm (mV)
Read Total Voltage Alarm	0x5A	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Total Overvoltage Level 1 Alarm (0.1V). Byte2-byte3: Total Overvoltage Level 2 Alarm (0.1V). Byte4-byte5: Total Undervoltage Level 1 Alarm (0.1V). Byte6-byte7: Total Undervoltage Level 2 Alarm (0.01V)
Read Current Alarm	0x5B	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Charging Overcurrent Level 1 Alarm (30000 Offset, 0.1A). Byte2-byte3: Charging Overcurrent Level 2 Alarm (30000 Offset, 0.1A). Byte4-byte5: Discharging Overcurrent Level 1 Alarm (30000 Offset, 0.1A). Byte6-byte7: Discharging Overcurrent Level 2 Alarm (30000 Offset, 0.1A)
Read Temperature Alarm	0x5C	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charging Overtemperature Level 1 Alarm (40 Offset, ° C). Byte1: Charging Overtemperature Level 2 Alarm (40 Offset, ° C). Byte2: Charging Undertemperature Level 1 Alarm (40 Offset, ° C). Byte3: Charging Undertemperature Level 2 Alarm (40 Offset, ° C). Byte4: Discharging Overtemperature Level 1 Alarm (40 Offset, ° C). Byte5: Discharging Overtemperature Level 2 Alarm (40 Offset, ° C). Byte6: Discharging Undertemperature Level 1 Alarm (40 Offset, ° C). Byte7: Discharging Undertemperature Level 2 Alarm (40 Offset, ° C)
Read SOC Parameters	0x5D	Send	Byte0~Byte7: Reserved
		Return	Byte0-Byte1: SOC Over Level 1 Alarm Value (0.1%). Byte2-Byte3: SOC Over Level 2 Alarm Value (0.1%).

			Byte4-Byte5: SOC Under Level 1 Alarm Value (0.1%). Byte6-Byte7: SOC Under Level 2 Alarm Value (0.1%)
Read Voltage/Temperature Difference Alarm	0x5E	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Voltage Difference Over Level 1 Alarm Value (mV). Byte2-byte3: Voltage Difference Over Level 2 Alarm Value (mV). Byte4: Temperature Difference Over Level 1 Alarm Value (° C). Byte5: Temperature Difference Over Level 2 Alarm Value (° C)
Read Balancing Parameters	0x5F	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Balancing Start Voltage (mV). Byte2-byte3: Balancing Start Voltage Difference (mV). Byte4-byte5: Byte6-byte7:
Read Current Parameters	0x60	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Current Alarm Value (A). Byte2-byte3: Current Sense Resistor (1μΩ)
Read RTC	0x61	Send	Byte0~Byte7:Reserved
		Return	Byte0: Year (2000 Offset). Byte1: Month. Byte2: Day. Byte3: Hour. Byte4: Minute. Byte5: Second
Software Version Number	0x62	Send	Byte0~Byte7:Reserved
		Return	14 bytes, sent in 2 frames. Byte0: Frame number, starting from 0. Byte1-byte7: Version number
Hardware Version Number	0x63	Send	Byte0~Byte7:Reserved
		Return	14 bytes, sent in 2 frames. Byte0: Frame number, starting from 0. Byte1-byte7: Version number
Fault Record	0x64	Send	The upper computer triggers reading all fault records. Only one message is sent, and it waits for reception to be complete. Byte0~Byte7: Reserved
		Return	Data parsing refers to Table 1, single fault record data structure. Byte0: Frame number (0x00-0x7F, 0xFF means all faults have been transmitted). Byte1-Byte7: Fault record data
Read Board Number	0x65	Send	Byte0~Byte7:Reserved
		Return	Byte 0: board number Byte1:sum bmu number

			Byte 2-Byte 7:Reserved
Read Heating Temperature	0x66	Send	Byte0~Byte7: Reserved
		Return	Byte 0: Heating Start Temperature (40 Offset, ° C). Byte 2: Fan Start Temperature (40 Offset, ° C). Byte4: Key Control MOS (0x55: No control, 0xA5: Control discharge MOS and sleep, 0x5A: Control discharge MOS but not sleep). Byte 5-Byte 7: Reserved
Read Active Balancing Switch	0x67	Send	Byte0~Byte7: Reserved
		Return	Byte 0: Active Balancing Switch (0: OFF, 1: ON). Byte 1-Byte 7: Reserved
Read Active Balancing Parameters	0x68	Send	Byte0~Byte7: Reserved
		Return	Byte 0: Number of battery cells in series. Byte 1-Byte 2: Balancing Current (0.1A). Byte 3-Byte 7: Reserved
Read Inverter Parameters	0x69	Send	Byte0~Byte7: Reserved
		Return	Byte 0: Inverter Type (0: NONE, 1: PYLON, 2: GROWAT, 3: SOFAR, 4: VOLTRONICPOWER, 5: GOODWE, 6: SRNE, 7: MUST, 8: VICTRONENERGY). Byte 1: Inverter Communication Type (0: 485 communication, 1: CAN communication). Byte 2-Byte 7: Reserved
Read SN Serial Number	0x6A	Send	Byte0~Byte7: Reserved
		Return	32 bytes, sent in 5 frames. Byte0: Frame number, starting from 1. Byte1-byte7: data
Read Cell Overvoltage Level 1 Alarm	0x70	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Cell Overvoltage Level 1 Alarm Value (mV). Byte2-byte3: Cell Overvoltage Level 1 Alarm Delay (0.01S). Byte4-byte5: Cell Overvoltage Level 1 Alarm Recovery Value (mV). Byte6-byte7: Cell Overvoltage Level 1 Alarm Recovery Delay (0.01S)
Read Cell Undervoltage Level 1 Alarm	0x71	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Cell Undervoltage Level 1 Alarm Value (mV). Byte2-byte3: Cell Undervoltage Level 1 Alarm Delay (0.01S). Byte4-byte5: Cell Undervoltage Level 1 Alarm Recovery Value (mV). Byte6-byte7: Cell Undervoltage Level 1 Alarm Recovery Delay (0.01S)
Read Current Level 1 Alarm	0x72	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Discharge Overcurrent Level 1 Alarm Value (30000 Offset, 0.1A). Byte2-byte3: Discharge Overcurrent Level 1 Alarm Delay (0.01S). Byte4-byte5: Charge Overcurrent Level 1 Alarm Value (30000 Offset, 0.1A). Byte6-byte7: Charge Overcurrent Level 1 Alarm Delay (0.01S)

Read High Temperature Level 1 Alarm	0x73	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charging Overtemperature Level 1 Alarm (40 Offset, ° C). Byte1: Charging Overtemperature Level 1 Alarm Delay (0.1S). Byte2: Charging Overtemperature Level 1 Alarm Recovery Value (40 Offset, ° C). Byte3: Charging Overtemperature Level 1 Alarm Recovery Delay (0.1S). Byte4: Discharging Overtemperature Level 1 Alarm (40 Offset, ° C). Byte5: Discharging Overtemperature Level 1 Alarm Delay (0.1S). Byte6: Discharging Overtemperature Level 1 Alarm Recovery Value (40 Offset, ° C). Byte7: Discharging Overtemperature Level 1 Alarm Recovery Delay (0.1S)
Read Low Temperature Level 1 Alarm	0x74	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charging Undertemperature Level 1 Alarm (40 Offset, ° C). Byte1: Charging Undertemperature Level 1 Alarm Delay (0.1S). Byte2: Charging Undertemperature Level 1 Alarm Recovery Value (40 Offset, ° C). Byte3: Charging Undertemperature Level 1 Alarm Recovery Delay (0.1S). Byte4: Discharging Undertemperature Level 1 Alarm (40 Offset, ° C). Byte5: Discharging Undertemperature Level 1 Alarm Delay (0.1S). Byte6: Discharging Undertemperature Level 1 Alarm Recovery Value (40 Offset, ° C). Byte7: Discharging Undertemperature Level 1 Alarm Recovery Delay (0.1S)
Read Total Overvoltage Level 1 Alarm	0x75	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Total Overvoltage Level 1 Alarm Value (0.1V). Byte2-byte3: Total Overvoltage Level 1 Alarm Delay (0.01S). Byte4-byte5: Total Overvoltage Level 1 Alarm Recovery Value (0.1V). Byte6-byte7: Total Overvoltage Level 1 Alarm Recovery Delay (0.01S)
Read Total Undervoltage Level 1 Alarm	0x76	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Total Undervoltage Level 1 Alarm Value (0.1V). Byte2-byte3: Total Undervoltage Level 1 Alarm Delay (0.01S). Byte4-byte5: Total Undervoltage Level 1 Alarm Recovery Value (0.1V). Byte6-byte7: Total Undervoltage Level 1 Alarm Recovery Delay (0.01S)
Read Voltage Difference Level 1 Alarm	0x77	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Voltage Difference Over Level 1 Alarm Value (1mV). Byte2-byte3: Voltage Difference Over Level 1 Alarm Delay (0.01S). Byte4-byte5: Voltage Difference Over Level 1 Alarm Recovery Value (0.1V). Byte6-byte7: Voltage Difference Over Level 1 Alarm Recovery Delay (0.01S)
Read Temperature Difference Level 1 Alarm	0x78	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Temperature Difference Over Level 1 Alarm Value (° C). Byte2-byte3: Temperature Difference Over Level 1 Alarm Delay (0.01S). Byte4-byte5: Temperature Difference Over Level 1 Alarm Recovery Value (0.1V). Byte6-byte7: Temperature Difference Over Level 1 Alarm Recovery Delay (0.01S)

Read Cell Overvoltage Level 2 Alarm	0x79	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Cell Overvoltage Level 2 Alarm Value (mV). Byte2-byte3: Cell Overvoltage Level 2 Alarm Delay (0.01S). Byte4-byte5: Cell Overvoltage Level 2 Alarm Recovery Value (mV). Byte6-byte7: Cell Overvoltage Level 2 Alarm Recovery Delay (0.01S)
Read Cell Undervoltage Level 2 Alarm	0x7A	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Cell Undervoltage Level 2 Alarm Value (mV). Byte2-byte3: Cell Undervoltage Level 2 Alarm Delay (0.01S). Byte4-byte5: Cell Undervoltage Level 2 Alarm Recovery Value (mV). Byte6-byte7: Cell Undervoltage Level 2 Alarm Recovery Delay (0.01S)
Read Current Level 2 Alarm	0x7B	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Discharge Overcurrent Level 2 Alarm Value (30000 Offset, 0.1A). Byte2-byte3: Discharge Overcurrent Level 2 Alarm Delay (0.01S). Byte4-byte5: Charge Overcurrent Level 2 Alarm Value (30000 Offset, 0.1A). Byte6-byte7: Charge Overcurrent Level 2 Alarm Delay (0.01S)
Read High Temperature Level 2 Alarm	0x7C	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charging Overtemperature Level 2 Alarm (40 Offset, ° C). Byte1: Charging Overtemperature Level 2 Alarm Delay (0.1S). Byte2: Charging Overtemperature Level 2 Alarm Recovery Value (40 Offset, ° C). Byte3: Charging Overtemperature Level 2 Alarm Recovery Delay (0.1S). Byte4: Discharging Overtemperature Level 2 Alarm (40 Offset, ° C). Byte5: Discharging Overtemperature Level 2 Alarm Delay (0.1S). Byte6: Discharging Overtemperature Level 2 Alarm Recovery Value (40 Offset, ° C). Byte7: Discharging Overtemperature Level 2 Alarm Recovery Delay (0.1S)
Read Low Temperature Level 2 Alarm	0x7D	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charging Undertemperature Level 2 Alarm (40 Offset, ° C). Byte1: Charging Undertemperature Level 2 Alarm Delay (0.1S). Byte2: Charging Undertemperature Level 2 Alarm Recovery Value (40 Offset, ° C). Byte3: Charging Undertemperature Level 2 Alarm Recovery Delay (0.1S). Byte4: Discharging Undertemperature Level 2 Alarm (40 Offset, ° C). Byte5: Discharging Undertemperature Level 2 Alarm Delay (0.1S). Byte6: Discharging Undertemperature Level 2 Alarm Recovery Value (40 Offset, ° C). Byte7: Discharging Undertemperature Level 2 Alarm Recovery Delay (0.1S)
Read Total Overvoltage Level 2 Alarm	0x7E	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Total Overvoltage Level 2 Alarm Value (0.1V). Byte2-byte3: Total Overvoltage Level 2 Alarm Delay (0.01S). Byte4-byte5: Total Overvoltage Level 2 Alarm Recovery Value (0.1V). Byte6-byte7: Total Overvoltage Level 2 Alarm Recovery Delay (0.01S)

Read Total Undervoltage Level 2 Alarm	0x7F	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Total Undervoltage Level 2 Alarm Value (0.1V). Byte2-byte3: Total Undervoltage Level 2 Alarm Delay (0.01S). Byte4-byte5: Total Undervoltage Level 2 Alarm Recovery Value (0.1V). Byte6-byte7: Total Undervoltage Level 2 Alarm Recovery Delay (0.01S)
Read Voltage Difference Level 2 Alarm	0x80	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Voltage Difference Over Level 2 Alarm Value (1mV). Byte2-byte3: Voltage Difference Over Level 2 Alarm Delay (0.01S). Byte4-byte5: Voltage Difference Over Level 2 Alarm Recovery Value (0.1V). Byte6-byte7: Voltage Difference Over Level 2 Alarm Recovery Delay (0.01S)
Read Temperature Difference Level 2 Alarm	0x81	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Temperature Difference Over Level 2 Alarm Value (° C). Byte2-byte3: Temperature Difference Over Level 2 Alarm Delay (0.01S). Byte4-byte5: Temperature Difference Over Level 2 Alarm Recovery Value (0.1V). Byte6-byte7: Temperature Difference Over Level 2 Alarm Recovery Delay (0.01S)
Read Cell Overvoltage Level 3 Alarm	0x82	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Cell Overvoltage Level 3 Alarm Value (mV). Byte2-byte3: Cell Overvoltage Level 3 Alarm Delay (0.01S). Byte4-byte5: Cell Overvoltage Level 3 Alarm Recovery Value (mV). Byte6-byte7: Cell Overvoltage Level 3 Alarm Recovery Delay (0.01S)
Read Cell Undervoltage Level 3 Alarm	0x83	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Cell Undervoltage Level 3 Alarm Value (mV). Byte2-byte3: Cell Undervoltage Level 3 Alarm Delay (0.01S). Byte4-byte5: Cell Undervoltage Level 3 Alarm Recovery Value (mV). Byte6-byte7: Cell Undervoltage Level 3 Alarm Recovery Delay (0.01S)
Read Current Level 3 Alarm	0x84	Send	Byte0~Byte7: Reserved
		Return	Byte0-byte1: Discharge Overcurrent Level 3 Alarm Value (30000 Offset, 0.1A). Byte2-byte3: Discharge Overcurrent Level 3 Alarm Delay (0.01S). Byte4-byte5: Charge Overcurrent Level 3 Alarm Value (30000 Offset, 0.1A). Byte6-byte7: Charge Overcurrent Level 3 Alarm Delay (0.01S)
Read High Temperature Level 3 Alarm	0x85	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charging Overtemperature Level 3 Alarm (40 Offset, ° C). Byte1: Charging Overtemperature Level 3 Alarm Delay (0.1S). Byte2: Charging Overtemperature Level 3 Alarm Recovery Value (40 Offset, ° C). Byte3: Charging Overtemperature Level 3 Alarm Recovery Delay (0.1S). Byte4: Discharging Overtemperature Level 3 Alarm (40 Offset, ° C). Byte5: Discharging Overtemperature Level 3 Alarm Delay (0.1S). Byte6: Discharging Overtemperature Level 3 Alarm Recovery Value (40 Offset, ° C).

Read Low Temperature Level 3 Alarm	0x86	Send	Byte7: Discharging Overtemperature Level 3 Alarm Recovery Delay (0.1S) Byte0~Byte7: Reserved
		Return	Byte0: Charging Undertemperature Level 3 Alarm (40 Offset, ° C). Byte1: Charging Undertemperature Level 3 Alarm Delay (0.1S). Byte2: Charging Undertemperature Level 3 Alarm Recovery Value (40 Offset, ° C). Byte3: Charging Undertemperature Level 3 Alarm Recovery Delay (0.1S). Byte4: Discharging Undertemperature Level 3 Alarm (40 Offset, ° C). Byte5: Discharging Undertemperature Level 3 Alarm Delay (0.1S). Byte6: Discharging Undertemperature Level 3 Alarm Recovery Value (40 Offset, ° C). Byte7: Discharging Undertemperature Level 3 Alarm Recovery Delay (0.1S)
Read Total Overvoltage Level 3 Alarm	0x87	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte1: Total Overvoltage Level 3 Alarm Value (0.1V). Byte2~byte3: Total Overvoltage Level 3 Alarm Delay (0.01S). Byte4~byte5: Total Overvoltage Level 3 Alarm Recovery Value (0.1V). Byte6~byte7: Total Overvoltage Level 3 Alarm Recovery Delay (0.01S)
Read Total Undervoltage Level 3 Alarm	0x88	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte1: Total Undervoltage Level 3 Alarm Value (0.1V). Byte2~byte3: Total Undervoltage Level 3 Alarm Delay (0.01S). Byte4~byte5: Total Undervoltage Level 3 Alarm Recovery Value (0.1V). Byte6~byte7: Total Undervoltage Level 3 Alarm Recovery Delay (0.01S).
Read Voltage Difference Level 3 Alarm	0x89	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte1: Voltage Difference Over Level 3 Alarm Value (1mV). Byte2~byte3: Voltage Difference Over Level 3 Alarm Delay (0.01S). Byte4~byte5: Voltage Difference Over Level 3 Alarm Recovery Value (0.1V). Byte6~byte7: Voltage Difference Over Level 3 Alarm Recovery Delay (0.01S).
Read Temperature Difference Level 3 Alarm	0x8A	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte1: Temperature Difference Over Level 3 Alarm Value (° C). Byte2~byte3: Temperature Difference Over Level 3 Alarm Delay (0.01S). Byte4~byte5: Temperature Difference Over Level 3 Alarm Recovery Value (0.1V). Byte6~byte7: Temperature Difference Over Level 3 Alarm Recovery Delay (0.01S).
Total Voltage, Current, SOC	0x90	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte1: Cumulative Total Voltage (0.1V). Byte2~byte3: Acquired Total Voltage (0.1V). Byte4~byte5: Current (30000 Offset, 0.1A). Byte6~Byte7: SOC (0.1%).
Individual Cell Highest/Lowest	0x91	Send	Byte0~Byte7: Reserved
		Return	Byte0~byte1: Highest Cell Voltage Value (mV). Byte2: Highest Cell Voltage cell number.

Voltage			Byte3~byte4: Lowest Cell Voltage Value (mV). Byte5: Lowest Cell Voltage cell number.
Individual Cell Highest/Lowest Temperature	0x92	Send	Byte0~Byte7: Reserved
		Return	Byte0: Highest Cell Temperature Value (40 Offset, ° C). Byte1: Highest Cell Temperature cell number. Byte2: Lowest Cell Temperature Value (40 Offset, ° C). Byte3: Lowest Cell Temperature cell number.
Charge/Discharge, MOS Status	0x93	Send	Byte0~Byte7: Reserved
		Return	Byte0: Charge/Discharge Status (0: Idle, 1: Charging, 2: Discharging). Byte1: Charge MOS state. Byte2: Discharge MOS state. Byte3: BMS life (0~255 cycle). Byte4~Byte7: Remaining capacity (mAh).
Status Information 1	0x94	Send	Byte0~Byte7: Reserved
		Return	Byte0: Number of cells in series. Byte1: Number of temperature sensors. Byte2: Charger status (0: Disconnected, 1: Connected). Byte3: Load status (0: Disconnected, 1: Connected). Byte4: Bit 0: DI1 state. Bit 1: DI2 state. Bit 2: DI3 state. Bit 3: DI4 state. Bit 4: DO1 state. Bit 5: DO2 state. Bit 6: DO3 state. Bit 7: DO4 state. Byte5~Byte6: Charge/Discharge cycle count. Byte7: On-board temperature (40 Offset, ° C).
Individual Cell Voltage 1~48	0x95	Send	Byte0~Byte7: Reserved
		Return	Each cell voltage occupies 2 bytes, sent according to the actual number of cells, max 96 bytes, sent in 16 frames. Byte0: Frame number, starting from 0, 0xFF is invalid. Byte1~byte6: Cell voltage (1mV). Byte7: Reserved.
Individual Cell Temperature 1~16	0x96	Send	Byte0~Byte7:Reserved
		Return	Each temperature occupies 1 byte, sent according to the actual number of temperatures used, max 21 bytes, sent in 3 frames. Byte0: Frame number, starting from 0. Byte1~byte7: Cell temperature (40 Offset, ° C).
Individual Cell Balancing Status	0x97	Send	Byte0~Byte7:Reserved
		Return	0: OFF, 1: ON. Bit0: Cell 1 balancing status. ... Bit47: Cell 48 balancing status. Bit48~Bit63: Reserved.
Battery Fault Status	0x98	Send	Byte0~Byte7:Reserved
		Return	0 -> No error 1 -> Error

			Byte 0: Bit 0: Cell Overvoltage Level 1 Alarm. Bit 1: Cell Overvoltage Level 2 Alarm. Bit 2: Cell Undervoltage Level 1 Alarm. Bit 3: Cell Undervoltage Level 2 Alarm. Bit 4: Total Overvoltage Level 1 Alarm. Bit 5: Total Overvoltage Level 2 Alarm. Bit 6: Total Undervoltage Level 1 Alarm. Bit 7: Total Undervoltage Level 2 Alarm. Byte 1: Bit 0: Charging Overtemperature Level 1 Alarm. Bit 1: Charging Overtemperature Level 2 Alarm. Bit 2: Charging Undertemperature Level 1 Alarm. Bit 3: Charging Undertemperature Level 2 Alarm. Bit 4: Discharging Overtemperature Level 1 Alarm. Bit 5: Discharging Overtemperature Level 2 Alarm. Bit 6: Discharging Undertemperature Level 1 Alarm. Bit 7: Discharging Undertemperature Level 2 Alarm. Byte 2: Bit 0: Charge Overcurrent Level 1 Alarm. Bit 1: Charge Overcurrent Level 2 Alarm. Bit 2: Discharge Overcurrent Level 1 Alarm. Bit 3: Discharge Overcurrent Level 2 Alarm. Bit 4: SOC Over Level 1 Alarm. Bit 5: SOC Over Level 2 Alarm. Bit 6: SOC Under Level 1 Alarm. Bit 7: SOC Under Level 2 Alarm. Byte 3: Bit 0: Voltage Difference Over Level 1 Alarm. Bit 1: Voltage Difference Over Level 2 Alarm. Bit 2: Temperature Difference Over Level 1 Alarm. Bit 3: Temperature Difference Over Level 2 Alarm. Bit 4~Bit7: Reserved. Byte 4: Bit 0: Charge MOS Overtemperature Warning. Bit 1: Discharge MOS Overtemperature Warning. Bit 2: Charge MOS Temperature Sensor Fault. Bit 3: Discharge MOS Temperature Sensor Fault. Bit 4: Charge MOS Adhesion Fault. Bit 5: Discharge MOS Adhesion Fault.
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			Bit 6: Charge MOS Open Circuit Fault. Bit 7: Discharge MOS Open Circuit Fault. Byte 5: Bit 0: AFE Acquisition Chip Fault. Bit 1: Cell Acquisition Dropout. Bit 2: Cell Temperature Sensor Fault. Bit 3: EEPROM Storage Fault. Bit 4: RTC Clock Fault. Bit 5: Pre-charge Failure. Bit 6: Vehicle Communication Fault. Bit 7: Internal Network Communication Module Fault. Byte 6: Bit 0: Current Module Fault. Bit 1: Internal Total Voltage Detection Module Fault. Bit 2: Short Circuit Protection Fault. Bit 3: Low Voltage Disables Charging Fault. Bit 4: GPS or Soft Switch Disconnects MOS. Bit 5~Bit7: Reserved. Byte 7: Fault Code (e.g., 0x03, then "Fault Code 3" is displayed, 0 is not displayed).
Individual Cell Balancing Status	0x99	Send	Byte0~Byte7: Reserved
		Return	Byte 0: Balancing Status (0: close, 1: open). Byte1~byte2: Balancing Current (30000 Offset, 0.1A). Byte3: Balancing Position. Byte 4-Byte 7: Reserved.
Battery Status	0x9A	Send	Byte0~Byte7: Reserved
		Return	Byte0-Byte1: SOH (0.1%) Byte 2-Byte 7: Reserved
Wake-up Source	0xD8	Send	Byte 0~Byte 7:Reserved
		Return	Byte0: Key Signal Wake-up. Byte1: Button Signal Wake-up. Byte2: 485 Signal Wake-up. Byte3: CAN Signal Wake-up. Byte4: Charge/Discharge Current Wake-up. Byte5~7: Reserved.
Discharge MOS Control	0xD9	Send	Byte 0: result (0 OFF, 1 ON) Byte 1-Byte 7: Reserved
		Return	Byte 0: result (0 OFF, 1 ON) Byte 1-Byte 7: Reserved

Charge MOS Control	0xDA	Send	Byte 0: result (0 OFF, 1 ON) Byte 1-Byte 7: Reserved
		Return	Byte 0: result (0 OFF, 1 ON) Byte 1-Byte 7: Reserved

Table 1: Single Fault Record Data Structure

Array Offset	Physical Quantity	Notes
0x00	Record Time Year	2000 Offset
0x01	Record Time Month	
0x02	Record Time Day	
0x03	Record Time Hour	
0x04	Record Time Minute	
0x05	Record Time Second	
0x06	Record ID	01: Start Charging, 02: End Charging, 03: Cell Overvoltage, 04: Cell Undervoltage, 05: Total Overvoltage, 06: Total Undervoltage, 07: Overtemperature, 08: Undertemperature, 09: Overcurrent, Other fault IDs are reserved.
0x07	Occurred or Cleared	0: Cleared, 1: Occurred
0x08~0x09	Battery Pack Total Voltage	0.1V
0x0A~0x0B	Battery Output Current	30000 Offset, 0.1A

0x0C~0x0D	SOC	0.1%
0x0E	MOS 状态	Bit0: Charge MOS State, Bit1: Discharge MOS State, Bit2~bit7: Reserved
0x0F~0x10	Highest Cell Voltage	mV
0x11	Highest Cell Voltage Number	
0x12~0x13	Lowest Cell Voltage	mV
0x14	Lowest Cell Voltage Number	
0x15	Highest Cell Temperature	40 Offset, °C
0x16	Highest Cell Temperature Number	
0x17	Lowest Cell Temperature	40 Offset, °C
0x18	Lowest Cell Temperature Number	
0x19	Fault Code	Reserved
0x1A	CRC	Sum of bytes 0-17